Pediatric Tachycardia (<13 years)

Initial Assessment & Support
- Maintain patent airway, assist breathing as necessary
- Administer oxygen as required
- Cardiac monitoring to identify rhythm, monitor pulse, BP and Sp02

Evaluate 12-Lead ECG

Probable Sinus Tachycardia if:
- P waves are present/normal
- Variable R-R Interval
- Infant rate <220/min
- Child rate usually <180bpm

Identify and Treat underlying cause

Cardiopulmonary Compromise?
- Acutely altered mental status
- Signs of Shock
- Hypotension

YES

Evaluate QRS Duration

Narrow ≤0.09sec

Probable SVT if:
- P waves absent
- R-R interval not variable
- Infant usually >220
- Child usually >180
- History of abrupt rate change

Adenosine IV/IO
1st Dose: 0.1mg/kg
2nd Dose: 0.2mg/kg

P

Synchronized Cardioversion
0.5-1J/kg
Then **2J/kg**

AP

Wide ≥0.09sec

Possible VT

Synchronized Cardioversion
0.5-1J/kg
Then **2J/kg**

AP

NO

Evaluate QRS Duration

Narrow ≤0.09sec

Probable SVT if:
- P waves absent
- R-R interval not variable
- Infant usually >220
- Child usually >180
- History of abrupt rate change

Vagal Maneuvers

Adenosine IV/IO
1st Dose: 0.1mg/kg
2nd Dose: 0.2mg/kg

P

Wide ≥0.09sec

Possible VT

Consider Tele EMS

If rhythm is regular and monomorphic:

Adenosine IV/IO
1st Dose: 0.1mg/kg
2nd Dose: 0.2mg/kg

AP

Patient Care Goals

1. Rapidly identify tachycardia in infants and children and determine whether it is compromising perfusion or mental status.
2. Distinguish between physiologic sinus tachycardia (often a response to stress, pain or fever) and dangerous tachyarrhythmias requiring intervention.
3. Maintain adequate oxygenation, perfusion, and cerebral function.
4. Systematically explore and correct potential underlying causes, such as:
    - Excessive stimulants or medications (caffeine, decongestants, thyroid preparations, diet aids)
    - Illicit drug use (e.g., amphetamines, cocaine)
    - Pre-existing cardiac conditions (history of arrhythmias or structural heart disease)
    - Cardiac failure (e.g., CHF presenting as low-output tachycardia)
    - Severe infections (e.g., sepsis, pneumonia, meningitis)

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Infants and children ≤13 years presenting with a heart rate above normal for age, with or without signs of instability.

Age Group	Normal HR (awake)	Concerning HR (suggests pathological)
Infant (<1 yr)	100–160 bpm	>220 bpm
Child (1–10 yrs)	70–150 bpm	>180 bpm
Adolescent (>10 yrs)	60–100 bpm	>180 bpm

American Heart Association Pediatric Advanced Life Support Guidelines, 2020.

- Infants and children ≤13 years presenting with a heart rate above normal for age, with signs of poor perfusion, such as:
    - Altered mental status.
    - Weak pulses, hypotension, prolonged capillary refill (>3 seconds).
    - Pallor, cyanosis, respiratory distress.

Exclusion Criteria

- Age-appropriate sinus tachycardia (e.g., due to fever, pain, dehydration) without instability.
- Known chronic arrhythmias in stable condition.

Key Patient Assessment Points

1. **Primary Survey**
    - **Initial Impression**
        - Use the Pediatric Assessment Triangle (PAT) to assess appearance, work of breathing, circulation.
        - Look for evidence of shock or impending cardiac or respiratory arrest.
    - **Airway & Breathing Assessment**
        - Check for signs of respiratory distress, hypoxia, or failure.
        - Listen/Auscultate for wheezing, stridor, or silent chest.
    - **Circulation Assessment**
        - Assess central and peripheral pulses for rate, regularity, and strength.
        - Capillary refill time, skin color and temperature.
        - Blood pressure (if feasible).
2. **Secondary Survey**
    - **Heart Rate & Rhythm Assessment**
        - Count HR for a full minute.
        - Apply ECG if available to differentiate:
            - Sinus tachycardia (gradual onset, HR appropriate for age & context).

- Supraventricular tachycardia (SVT): abrupt onset, very high rate (>220 infant, >180 child), narrow QRS, poor perfusion.
- Ventricular tachycardia (VT): widened QRS.

3. **Reversible Cause Assessment**
   - Assess for Reversible causes such as: fever, dehydration, pain, hypoxia, anxiety, toxins, metabolic disorders.
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
   - Pediatric tachycardia should be triaged as:
     - Red (Serious & Life-threatening):
       - HR >220 in infants or >180 in children with signs of poor perfusion: altered mental status, hypotension, weak pulses, cyanosis, or respiratory distress.
       - Unstable tachycardia requiring immediate intervention (adenosine, cardioversion).
     - Yellow (Serious but not immediately life-threatening):
       - HR elevated beyond normal limits but with stable perfusion, normal mental status, and no signs of shock.
       - Patient responsive to initial vagal maneuvers, stable after intervention.

Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be routinely requested.
1. **Calm and systematic approach**
   - Keep the child warm and minimize stress.
   - Allow caregiver presence if calming.
2. **Oxygenation & Ventilation**
   - Ensure a patient airway is maintained
   - Administer Oxygen titrated to target oxygen saturation ≥94% and normocapnia ($EtCO_2$ 35–45 mmHg).
   - If respiratory failure or apnea, start bag-valve-mask ventilation.
3. **12-Lead ECG**
   - Perform a 12 lead ECG and continuously monitor the rhythm
4. **Measure Blood Glucose**
   - Check blood glucose and, if hypoglycemia is identified, treat according to the Hypoglycemia CPG; if hyperglycemia is identified, refer to the Hyperglycemia CPG for management.
5. **IV Access**
   - Establish IV access if indicated or in patients who:
     - Are at risk for clinical deterioration, or
     - Require IV fluid or medication administration.
   - Intraosseous (IO) access is indicated if IV access fails or is impractical.
6. **Cardiopulmonary Compromise Assessment**
   - **Stable**
     - Tachyarrhythmia with **adequate perfusion** (no significant hypotension, good mental status, warm extremities, normal capillary refill).
   - **Unstable**
     - Tachyarrhythmia with signs of **poor perfusion** or cardiovascular compromise, such as:
       - Altered mental status
       - Hypotension
       - Weak or absent peripheral pulses
       - Cool, mottled skin, prolonged capillary refill
       - Respiratory distress or failure
7. **Stable Tachycardia**
   - **Evaluate the rhythm**

- If **Narrow-Complex** (likely sinus tachycardia or SVT):
  - Assess underlying causes (such as: fever, pain, dehydration, anxiety).
  - If sinus tachycardia: treat underlying cause.
  - If SVT suspected:
    - **Attempt vagal maneuvers.**
    - If vagal maneuvers ineffective and IV/IO access available:
      - **Administer Adenosine:**
        - 1st dose: 0.1 mg/kg IV/IO rapid IV push, max 6 mg.
        - 2nd dose: 0.2 mg/kg IV/IO rapid IV push, max 12 mg.
- If **Wide-Complex**:
  - If rhythm is regular and monomorphic:
    - Adenosine may be administered after consultation with Tele EMS, routine administration is not recommended.
    - **Administer Adenosine:**
      - 1st dose: 0.1 mg/kg IV/IO rapid IV push, max 6 mg.
      - 2nd dose: 0.2 mg/kg IV/IO rapid IV push, max 12 mg.
  - If irregular or polymorphic → consult expert; avoid adenosine.
- Prepare for possible deterioration — have cardioversion and resuscitation equipment ready.

8. **Unstable Tachycardia (with a pulse)**
   - **Evaluate the Rhythm**
     - If **Narrow-Complex Tachycardia** (likely SVT)**:**
       - Consider **Vagal Maneuvers**
         - If clinical condition is sufficiently stable.
         - If vagal maneuvers are ineffective or the patient too unstable for vagal maneuvers:
       - Consider **Adenosine**
         - 1st dose: 0.1 mg/kg IV/IO rapid IV push, max 6 mg.
         - 2nd dose: 0.2 mg/kg IV/IO rapid IV push, max 12 mg.
         - If adenosine fails or the child deteriorates:
       - **Synchronized Cardioversion**
         - 1st Dose: 0.5–1 J/kg. If ineffective:
         - 2nd Dose: 2 J/kg.
     - If **Wide-Complex Tachycardia** (likely VT):
       - Proceed directly to **synchronized cardioversion**
         - 1st Dose: 0.5–1 J/kg, If ineffective:
         - 2nd Dose: 2 J/kg.
       - If rhythm is regular and monomorphic and child remains stable enough, consider adenosine.
       - Administer sedation if time and patient stability permits, but do not delay cardioversion.
9. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - Continuous ECG monitoring is required.
     - Assess for:
       - Arrhythmias
       - The response to any interventions (e.g., vagal maneuvers, adenosine, cardioversion), noting changes in rhythm and perfusion.

Patient Safety Considerations

- Avoid delays in cardioversion if the patient is unstable.
- Ensure correct dosing of adenosine and monitor for transient asystole.
- Be prepared to escalate to CPR if interventions fail.
- Minimize patient agitation to reduce cardiac demand.

Clinical Insights & Additional Guidance

- **Patient Stability & Action Table**

Feature	Stable	Unstable
Mental status	Alert, normal	Altered, confused, unresponsive
Perfusion	Good pulses, warm extremities, normal BP	Weak pulses, mottled/cool skin, hypotension
Action	Monitor, investigate & treat underlying cause	Immediate intervention: adenosine or synchronized cardioversion

Adapted from American Heart Association. Pediatric Advanced Life Support Guidelines. Circulation. 2020;142(16_suppl_2):S469–S523.

- Always consider that pediatric tachycardia is often secondary to hypoxia or shock — address underlying causes promptly.
- If rhythm is unclear, the patient deteriorates, or advanced interventions (e.g., cardioversion, higher-level airway management) may be needed,

Documentation

- Complete a comprehensive electronic Patient Care Record (ePCR), capturing the following elements to support continuity of care. Including:
  - Identification: Patient's name, age, sex, weight.
  - Medical complaint: Pediatric tachycardia — specify if stable or unstable, suspected type if known (e.g., SVT, VT).
  - Information/Injuries: Relevant history: congenital heart disease, recent infection, medications, previous episodes, any injuries.
  - Signs:
    - Heart rate and rhythm (trends and response to treatment)
    - Blood pressure, SpO_2, mental status, skin perfusion.
  - Treatment:
    - Oxygen and airway support
    - Vagal maneuvers performed
    - Medications (e.g., adenosine — dose & response)
    - Cardioversion (energy level & response)
  - Allergies: Any known allergies
  - Medications: Medications given at scene and any relevant home meds.
  - Background: Medical history, family history if relevant.
  - Time of onset, changes during care, and special considerations (e.g., ALS back-up requested, equipment used).
  - Other: Any other relevant information.

Key Performance Indicators

1. Perform an appropriate intervention (e.g., vagal maneuvers, adenosine, or cardioversion) for unstable tachycardia patients.

References

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